



Lemma

Look it up by looking.

A camera app that recognises everyday objects and teaches you their names in 12 languages — with pronunciation, grammar and colour.

Summer Internship · Mini Project

Rushan Haque · rushanhaque.online



RUNS ENTIRELY ON YOUR PHONE

THE PROBLEM

Learning words from a list doesn't stick

- **You memorise, then forget.** You learn “la voiture = car” on Monday and it is gone by Friday.
- **A word list has no context.** There is nothing for your brain to attach the word to.
- **You learn the wrong words.** Apps teach everyone the same 500 words — not the things actually in your room.
- **Grammar is left until later.** Most apps give you the word but not the gender, the article, or how it changes in a sentence.

“I know the word for this thing. I just can't remember it when I see the thing.”

THE IDEA

So point your camera at it instead

You see the object with your own eyes. You learn its name at that exact moment. Now your brain has something real to hang the word on.

1 Point

Aim the camera at anything — a chair, a cup, your dog.



2 It recognises

The app works out what the object is, right there on the phone.



3 It teaches

The word appears on screen with pronunciation, gender and colour.

No typing. No searching. No internet needed. Nothing leaves your phone.

What it looks like



Everything on one screen

- The word in big type, with its gender.
- How to say it, written simply.
- The colour of the object, in a full phrase.
- The English word underneath.
- Green brackets show what the app has locked on to.

Tap any label

Opens the full card: example sentences, your progress with that word, and the same word in the other 11 languages.

Four parts to the app

Camera

Point and learn. Quiz mode hides the answer until you tap. Switch camera, torch, zoom. Save a photo with the words on it. Or load a picture from your gallery.

Learn

21 decks of words. Five ways to be tested: remember it, pick it, hear it, fill the gap, type it. Search all 201 words.

Progress

Streaks, XP and levels. A 12-week activity chart. 21 achievements. Progress per language and per topic.

Settings

12 languages. Light and dark. Six colour themes. Detection tuning. Speech speed. Back up and restore everything.

The part I am most proud of

A camera can read one thing off an object that no word list can: its colour. So the app names the colour — and makes it agree with the noun, the way the language really works.

Point at a red car:

FRENCH

la voiture rouge

SPANISH

el coche rojo

GERMAN

das rote Auto

RUSSIAN

красная машина

JAPANESE

赤い車

Why this matters

The colour word changes shape depending on the object: rouge stays, rojo becomes roja, rot becomes rote. Getting that agreement right is the single hardest habit for a learner — and it is the one thing a camera can demonstrate directly, on a real object, in front of you.

THE PIPELINE

How it works, start to finish

This whole loop runs about eight times a second, inside the browser, on the phone.



SHADED = THE PARTS I WROTE MYSELF

The AI model is a ready-made one from Google. The interesting work is everything around it: making its output stable enough to put on screen, and turning a bare English label into a correct sentence in twelve languages.

What I used — the main pieces

TECHNOLOGY	WHAT IT IS	WHAT I USED IT FOR
TensorFlow.js	Google's AI library that runs inside a web	Running the object-detection model on the phone itself, with no
COCO-SSD	A ready-trained AI model that knows 80	Finding what is in the picture and where it is
JavaScript (ES Modules)	The language every browser speaks	All the app logic, split into 26 small, single-purpose files
HTML & CSS	The structure and the styling of a web	The whole interface, the design system, the animations, light and
Canvas API	A browser tool for drawing shapes and	Drawing the brackets around objects, and reading the colour off
Web Speech API	Text-to-speech built into the browser	Saying every word and sentence out loud in the right accent

What I used — the supporting tools

TOOL	WHAT IT IS	WHAT I USED IT FOR
localStorage	A small storage box the browser gives	Remembering your words, streak, settings and review schedule
Service Worker	A script that runs in the background, even	Making the app open and work with no internet after the first visit
PWA manifest	A small settings file browsers read	Letting you install Lemma to your home screen like a normal app
Node.js	JavaScript running outside the browser	Running my 95 automated tests, and generating the app icon in
Git	Version control — a history of every	Tracking the work, with a clear message explaining each change

No frameworks, no build step, no server

The app is plain HTML, CSS and JavaScript files. You can put the folder on any web host and it runs. That was a deliberate choice: fewer moving parts to break.

Making a noisy AI model usable

Straight out of the box the model flickers, mislabels things, and loses track of them. Seven steps of my own sit between the model and the screen.

1 Shape check

A short, wide box is a bad “bottle”, whatever the model says.

2 Size check

Reject boxes far too big or too small for that kind of object.

3 Remove duplicates

Two boxes on the same thing become one.

4 Break ties

A phone and a remote cannot both claim the same rectangle.

5 Match to last frame

Each object keeps its own identity between frames.

6 Wait before showing

A label appears only after several frames agree.

7 Smooth and predict

Labels glide, and lead slightly when you pan the phone.

The result

Two cats side by side stay two labels, instead of collapsing into one.

I don't store sentences — I build them

Writing every sentence by hand would mean thousands of lines of text to get wrong. Instead I store the bare word plus its gender, and a small engine works out the rest.

WHAT I STORE

apple

fr: pomme

gender: feminine

Two facts. That is all.



WHAT THE ENGINE WORKS OUT

- la pomme · une pomme
- C'est une pomme. Je vois une pomme. Où est la pomme ?
- l'orange — it knows to contract before a vowel
- der / die / das, and how German changes in "I see ..."
- la pomme blanche — the adjective agrees, automatically

201 words × 12 languages × 5 sentence patterns = over 12,000 correct sentences

Not one of them written by hand.

REMEMBERING

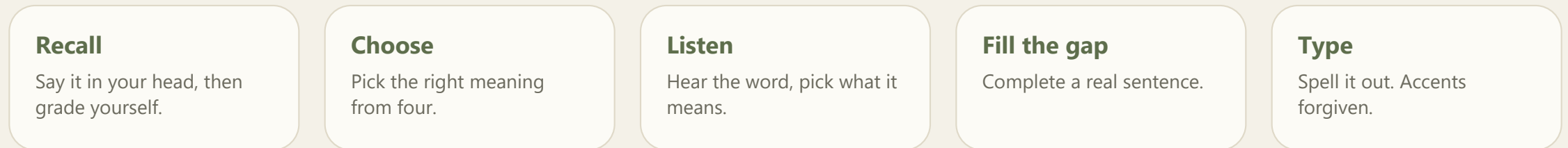
How it makes the words stick

Finding a word is easy. Keeping it is the hard part. Lemma uses spaced repetition — the same idea behind Anki — so you review each word just before you would have forgotten it.

Get it right, and the gap grows:



Five ways it tests you, so it never gets boring:



It never sends anything anywhere

A camera app that uploads what it sees is a privacy problem. Lemma does not have one, because there is nothing to upload to.

No server

There is no back end. The whole app is a folder of files.

No account

No sign-up, no email, no password, no tracking.

Nothing uploaded

Camera frames are read and thrown away in the same instant.

Works offline

After the first visit it runs with the internet switched off.

This also makes it very cheap to run and very hard to break: no database to maintain, no server bill, no user data to lose. For a project that has to keep working after the internship ends, that mattered.

The project in numbers

2,412

Translations

201 words × 12 languages

12

Languages

European, Asian and Arabic

21

Decks

12 by sight, 9 by topic

11

Colours

with full grammar agreement

~9,600

Lines of code

across 26 files

95

Automated tests

all passing

0

Dependencies

beyond the AI model

5

Review modes

plus quiz mode in camera

Built and tested end to end: camera, AI, grammar, review system, design and offline support.

Every piece written from scratch except the object-detection model itself.

Problems I hit, and how I fixed them

Two cats became one label

I was storing objects by their name, so a second cat overwrote the first. I rewrote it to give every object its own ID and follow it frame to frame.

Labels lagged when I moved the phone

The AI only runs a few times a second. I now measure each object's speed and predict where it will be, so the label keeps up.

Slow phones stuttered

I added a governor that times the AI and automatically lowers how often it runs, so the screen stays smooth instead of freezing.

The colour kept flickering red / orange

A colour is now only accepted after three frames in a row agree on it.

The app broke after an update

The offline cache was mixing old and new files. I made it serve one complete version at a time and only switch over when the user reloads.

People deny camera access

The app now explains why it needs the camera before the browser asks, instead of springing the prompt on you.

How I knew it actually worked

Pointing a phone at a chair is a slow way to check a bug is fixed. So I built ways to test it without the phone.

95 automated tests

They run in a couple of seconds and check the real logic: the review schedule, the grammar rules in all 12 languages, the object tracker, and that old saved progress still loads after an update.

A fake camera and a fake AI

I replaced the camera with a drawing of coloured blocks, and the AI with a stub that reports where they are. That let me test the whole pipeline instantly, and check colour detection with colours I chose.

Checked by hand in the browser

All 12 languages on screen, light and dark themes, phone and desktop layouts, and what happens when the camera is blocked or the model fails to download.

Several of the bugs on the previous slide were found by these tests, not by luck.

What I learned

- **Running AI in the browser is realistic now.** A phone can recognise objects in real time with no server behind it.
- **The model is only half the job.** Raw AI output is too noisy to show a user. Most of my effort went into making it steady enough to trust.
- **Design the data, not the feature.** Because words, languages and colours are all just data, adding a 13th language is a data job, not a rewrite.
- **Tests find what clicking around does not.** A whole class of bugs only appears when old saved data meets new code.
- **Privacy is a design decision.** Choosing to have no server removed a whole category of risk before it existed.
- **Small details decide whether people stay.** Explaining the camera prompt before it appears matters as much as the AI behind it.

What I would add next

1 Plural forms

So it can say “three apples” correctly. I left this out rather than risk teaching wrong grammar in German, Russian and Arabic.

2 More objects

A second model to go beyond the current 80 objects — ideally naming the breed of dog, not just “dog”.

3 Classroom mode

A teacher sets a list of objects; students go and find them. Turns the app into a scavenger hunt.

4 Better voices

Some phones have no voice installed for some languages. Bundling audio would fix that.

The app is finished and usable today — these are additions, not missing pieces.



Thank you

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Questions?